# Chapter 8. RNA Synthesis and Processing 2,3

## Lecture Contents

1. Introduction about learning performance of today's lecture
2. Transcription and its regulation in eukaryotes
3. Major RNA processing in eukaryotes - RNA editing
4. RNA turnover in eukaryotes

# 3. Major RNA Processing in eukaryotes (계속)

## • RNA editing

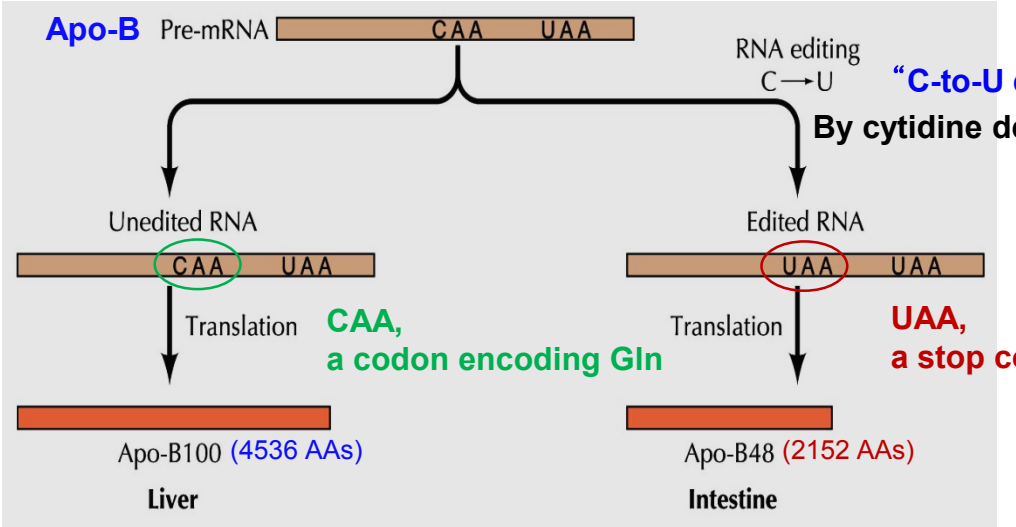
-Is **a processing that alters the protein-coding sequences of some mRNAs**.

-It involves **single base modification reactions** such as **deamination of cytosine to uridine and adenosine to inosine**.

-Editing of the mRNA for **apolipoprotein B (Apo-B)**, **which transports lipids in the blood**, results in two different proteins:

- **Apo-B100** is synthesized in **the liver** by translation of **the unedited mRNA**.
- **Apo-B48** is synthesized in **the intestine** from edited mRNA in which **a C has been changed to a U** by deamination.

**<Figure 8.54> Editing of apolipoprotein B mRNA**




Transport lipids in the circulation

Absorption of dietary lipids

**"Tissue-specific editing" → structurally and functionally different expression of apolipoprotein Bs !!**

# Chapter 8. RNA Synthesis and Processing 2,3

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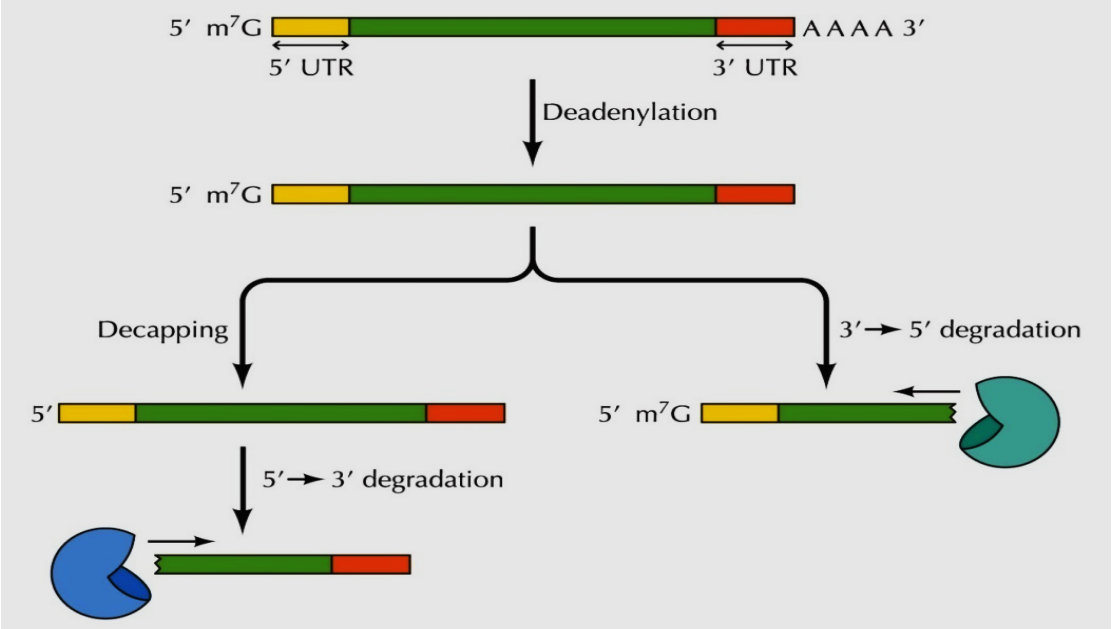
1. Introduction about learning performance of today's lecture
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4. **RNA turnover in eukaryotes**

# 4. RNA turnover in eukaryotes

## • mRNA degradation (turnover)

-Degradation of eukaryote mRNAs is initiated by shortening of poly-A tails.

**<Figure 8.55>**
**mRNA degradation**




-Rapidly degraded mRNAs often contain specific AU-rich sequences (ARE) near the 3′ ends which are binding sites for proteins that can either stabilize them or target them for degradation.

-The RNA-binding proteins (eg, HuR or tristetraproline (TTP)) are regulated by extracellular signals, such as growth factors and hormones.

-Degradation of some mRNAs is regulated by both siRNAs and miRNAs.

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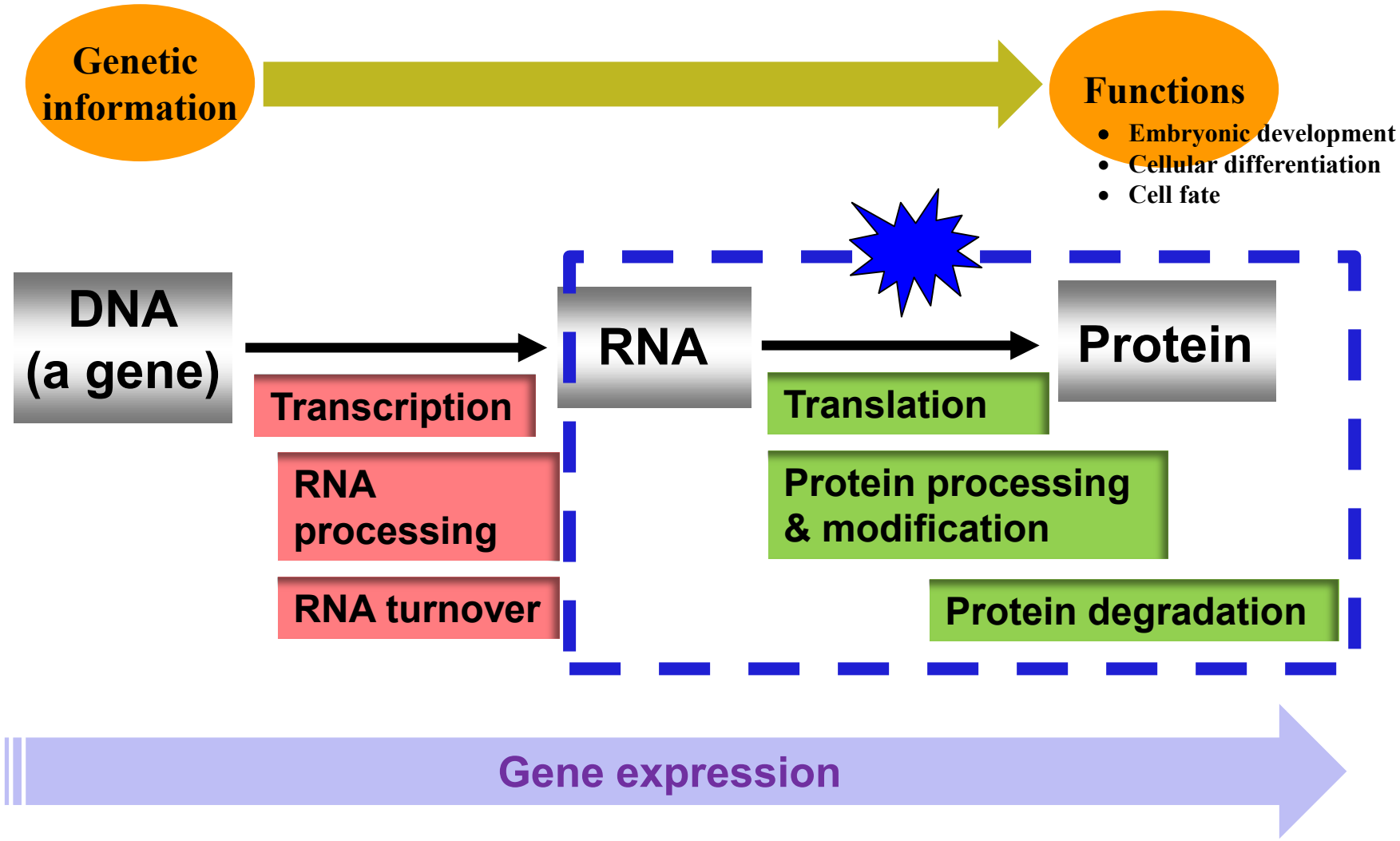
# 2026-2학기 세포의 분자적 이해

# Chapter 9
# Protein Synthesis, Processing, and Regulation (1)

분자의학교실 장병철

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# Chapter 9. Protein Synthesis, Processing, and Regulation (1)



# Lecture Contents

1. Introduction about learning performance of today's lecture
2. Translation (protein synthesis) of mRNA in eukaryotes
   - Overview of translation, codon, tRNA, ribosome
3. Three steps of translation in eukaryotes
   - Translation initiation, elongation, termination
   - Translation factors involved in each step
4. Regulation of translation in eukaryotes

# 1. Learning Performance

1) Be able to explain **translation (protein synthesis) in eukaryotic cells**.
2) Be able to describe **codons, tRNAs, and ribosomes** involved in eukaryotic translation.
3) Be able to explain **the stages of eukaryotic translation** (initiation, elongation, and termination).
4) Be able to describe **the translation factors** involved in initiation, elongation, and termination in eukaryotic cells.
5) Be able to explain **the regulatory mechanisms of translation** in eukaryotic cells.

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# Lecture Contents

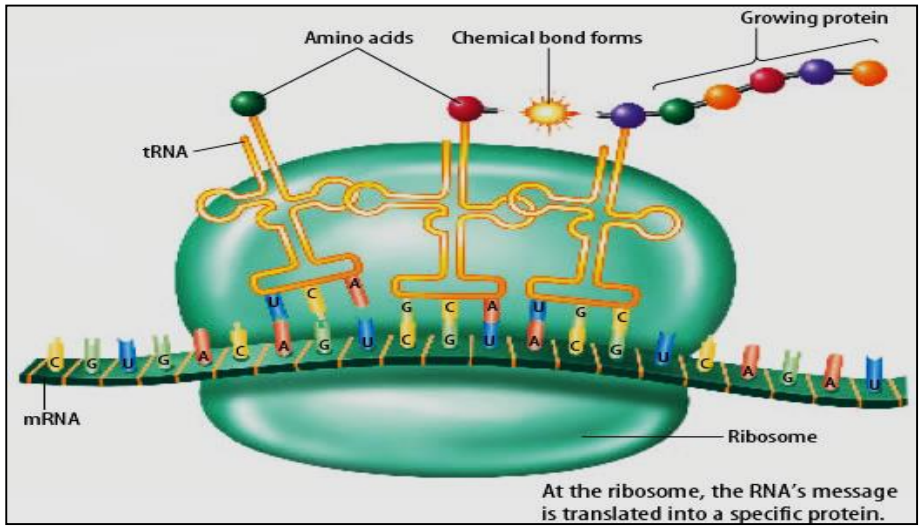
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# 2. Translation (protein synthesis) of mRNA in eukaryotes

## 2.1. Overview of translation

- Translation is **a process to synthesize proteins** (as directed by mRNA templates).
- It **occurs in the ribosome** of the endoplasmic reticulum (ER), particularly in rough ER.
- During translation, all mRNAs are read in the 5′ to 3′ direction, and **polypeptide chains are synthesized from the amino (N-) terminus to the carboxy (C-) terminus**.
- Each amino acid (AA) is specified by **three bases**, called **a codon**, on the (mature) mRNA.
- Translation is carried out on the ribosome, **with tRNAs serving as adaptors**.
- **Protein synthesis** in the ribosome involves **interactions between three types of RNA molecules (mRNA, tRNA, and rRNA) and many other proteins**.

5’




3’

# Lecture Contents

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# 2. Translation (protein synthesis) of mRNA in eukaryotes

## 2.2. Codon: Three bases on the mature mRNA, which can specify an amino acid (AA)

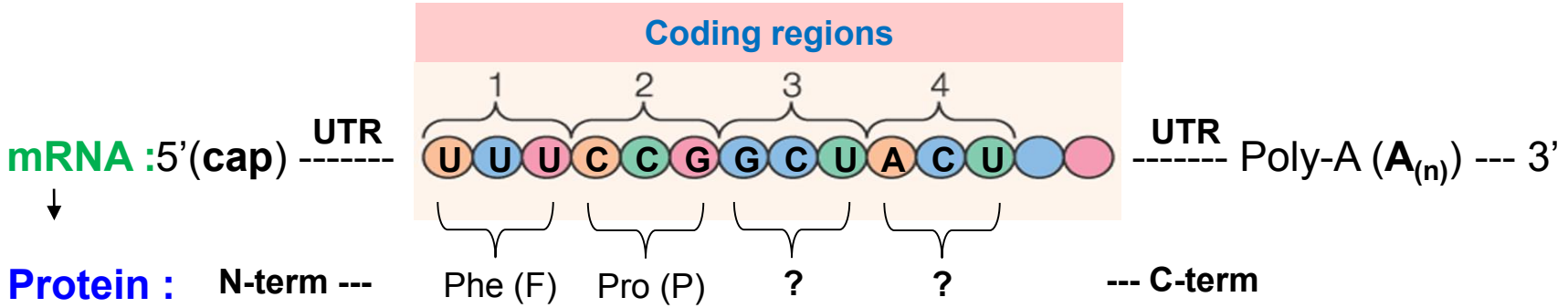



표 12.1 유전암호 (Genetic code) mRNA 상의 각 코돈 (codon)이 어떤 아미노산을 부호화(encoding) 할지를 정해 놓은 규칙.

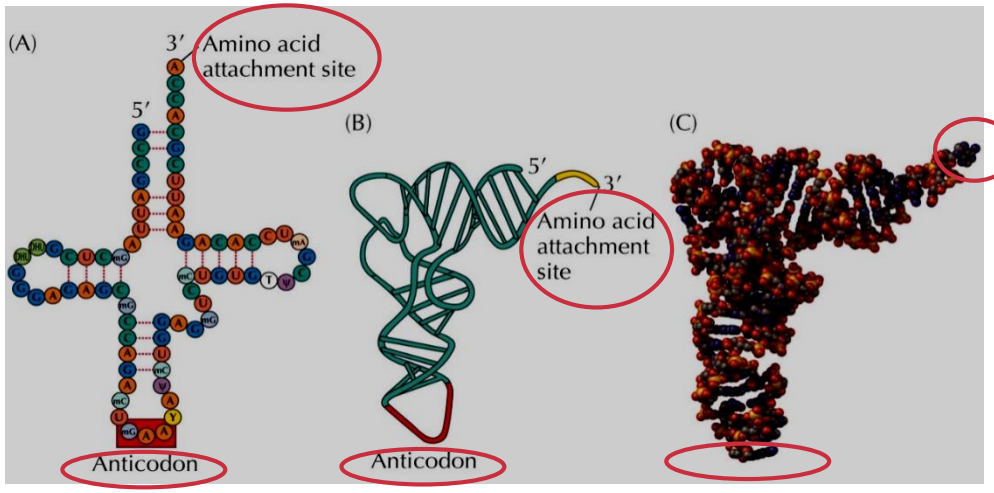
| 첫 번째 위치 (5′ 말단) | 두 번째 위치 | | | | 세 번째 위치 (3′ 말단) |
|---|---|---|---|---|---|
| | U | C | A | G | |
| U | UUU Phe | UCU Ser | UAU Tyr | UGU Cys | U |
| | UUC Phe | UCC Ser | UAC Tyr | UGC Cys | C |
| | UUA Leu | UCA Ser | UAA Stop | UGA Stop | A |
| | UUG Leu | UCG Ser | UAG Stop | UGG Trp | G |
| C | CUU Leu | CCU Pro | CAU His | CGU Arg | U |
| | CUC Leu | CCC Pro | CAC His | CGC Arg | C |
| | CUA Leu | CCA Pro | CAA Gln | CGA Arg | A |
| | CUG Leu | CCG Pro | CAG Gln | CGG Arg | G |
| A | AUU Ile | ACU Thr | AAU Asn | AGU Ser | U |
| | AUC Ile | ACC Thr | AAC Asn | AGC Ser | C |
| | AUA Ile | ACA Thr | AAA Lys | AGA Arg | A |
| | AUG Met* | ACG Thr | AAG Lys | AGG Arg | G |
| G | GUU Val | GCU Ala | GAU Asp | GGU Gly | U |
| | GUC Val | GCC Ala | GAC Asp | GGC Gly | C |
| | GUA Val | GCA Ala | GAA Glu | GGA Gly | A |
| | GUG Val | GCG Ala | GAG Glu | GGG Gly | G |

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# 2. Translation (protein synthesis) of mRNA in eukaryotes

## 2.3. tRNA



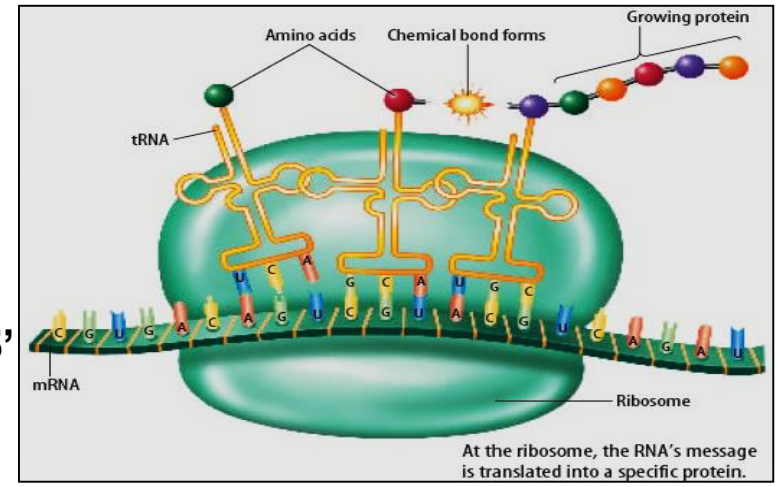




Figure 9.1. Structure of tRNAs

- **tRNAs align amino acids with corresponding codons on the mRNA template**.
- They are **70 to 80 nucleotides long** and have characteristic cloverleaf structures resulting from complementary base pairing between different regions.
- All tRNAs **have the sequence CCA at the 3′ terminus**, and **amino acids are covalently attached to the ribose of the terminal adenosine**.
- **The anticodon loop binds to the appropriate codon in the mRNA** by complementary base pairing.

표 12.1 유전암호

| 첫 번째 위치 (5′ 말단) | 두 번째 위치 | | | | 세 번째 위치 (3′ 말단) |
|---|---|---|---|---|---|
| | U | C | A | G | |
| U | UUU Phe | UCU Ser | UAU Tyr | UGU Cys | U |
| | UUC Phe | UCC Ser | UAC Tyr | UGC Cys | C |
| | UUA Leu | UCA Ser | UAA Stop | UGA Stop | A |
| | UUG Leu | UCG Ser | UAG Stop | UGG Trp | G |
| C | CUU Leu | CCU Pro | CAU His | CGU Arg | U |
| | CUC Leu | CCC Pro | CAC His | CGC Arg | C |
| | CUA Leu | CCA Pro | CAA Gln | CGA Arg | A |
| | CUG Leu | CCG Pro | CAG Gln | CGG Arg | G |
| A | AUU Ile | ACU Thr | AAU Asn | AGU Ser | U |
| | AUC Ile | ACC Thr | AAC Asn | AGC Ser | C |
| | AUA Ile | ACA Thr | AAA Lys | AGA Arg | A |
| | AUG Met* | ACG Thr | AAG Lys | AGG Arg | G |
| G | GUU Val | GCU Ala | GAU Asp | GGU Gly | U |
| | GUC Val | GCC Ala | GAC Asp | GGC Gly | C |
| | GUA Val | GCA Ala | GAA Glu | GGA Gly | A |
| | GUG Val | GCG Ala | GAG Glu | GGG Gly | G |

# 2. Translation (protein synthesis) of mRNA in eukaryotes

## 2.3. tRNA (계속)

- **Attachment of amino acid(s) to specific tRNAs** is mediated by **aminoacyl tRNA synthase (TAS)** **through two steps**.

▶ **Two steps by TAS**:

1) **The amino acid (AA) is joined to AMP**, forming **aminoacyl AMP** (“**activation of amino acid**”)

2) The aminoacyl AMP is then **transferred to the 3′ CCA terminus of the tRNA**, and AMP is released.

▶ **Each of these 20 TASs recognizes a single amino acid**, as well as **the correct tRNA to attach it to**.

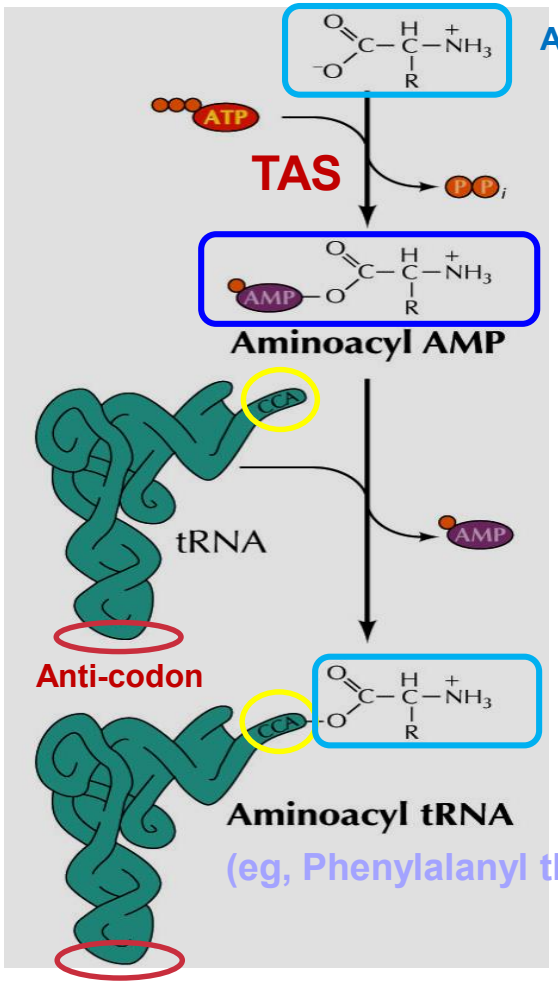



**Figure 9.2. Attachment of amino acid to tRNAs**

**Anti-codon**

- binds to the appropriate codon on the mRNA by complementary base pairing

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# 2. Translation (protein synthesis) of mRNA in eukaryotes

## 2.4. Ribosome

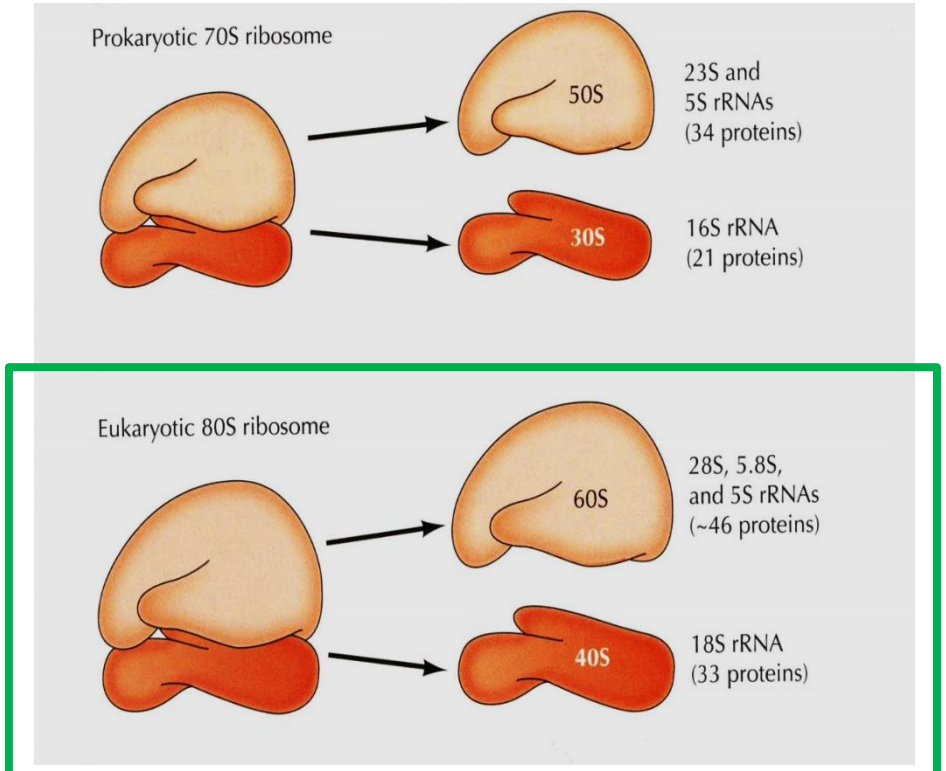



**Figure 9.4. Ribosome structure**

| 리보솜 단위 | rRNA 구성 | 단백질 수 | 주요 ribosomal proteins | 기능 |
|---|---|---|---|---|
| **40S 소단위** | 18S rRNA | ~33개 (eS 명명) | eS12, eS19, eS1, eS26 등 | mRNA decoding, 코돈-안티코돈 정확성 유지, mRNA 안정화 |
| **60S 대단위** | 28S, 5.8S, 5S rRNAs | ~47개 (eL/uL 명명) | eL6, eL14, eL22, eL23, uL3, uL4 등 | 펩티드 결합 형성 보조, 소단위 결합, nascent chain tunnel 안정화 |
| **특수 역할** | – | – | RPL5, RPL11, RPS3 등 | p53 경로, DNA repair, 세포 사멸 조절 등 번역 외 기능 |

Cells have many ribosomes:

- *E. coli* has about 20,000
- Growing mammalian cells have 10 million

▶ **Ribosomes** are very efficient organelles. A single ribosome in a eukaryotic cell can **add 2 amino acids to a protein chain every second**.

▶ **The units of a ribosome** are described by their **Svedberg (S)** values, which are based upon their rate of sedimentation in a centrifuge, depending on their size and shape. **The ribosomes in a eukaryotic cell generally have a Svedberg value of 80S** and are comprised of 40s and 60s subunits].

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# 3. Three steps of translation and translation factors

## 3.1. Translation initiation, elongation, termination

- **The 7-methylguanosine cap at the 5′ terminus of eukaryotic mRNAs** are **recognized by the 40S ribosomal subunit**.
- The 40S subunit then scans downstream of until they encounter **the translation initiation codon, AUG**.

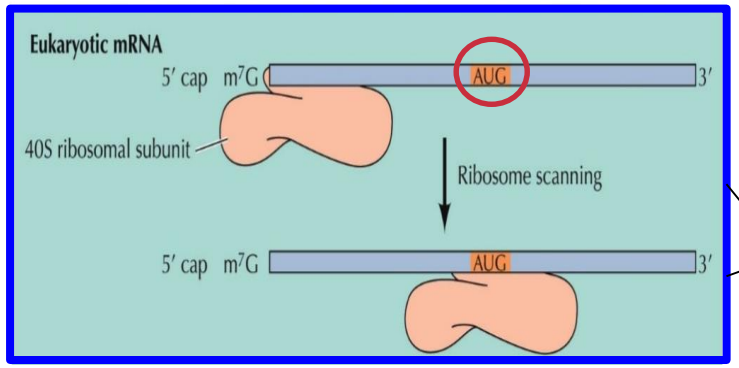



Figure 9.8. Signals for translation initiation

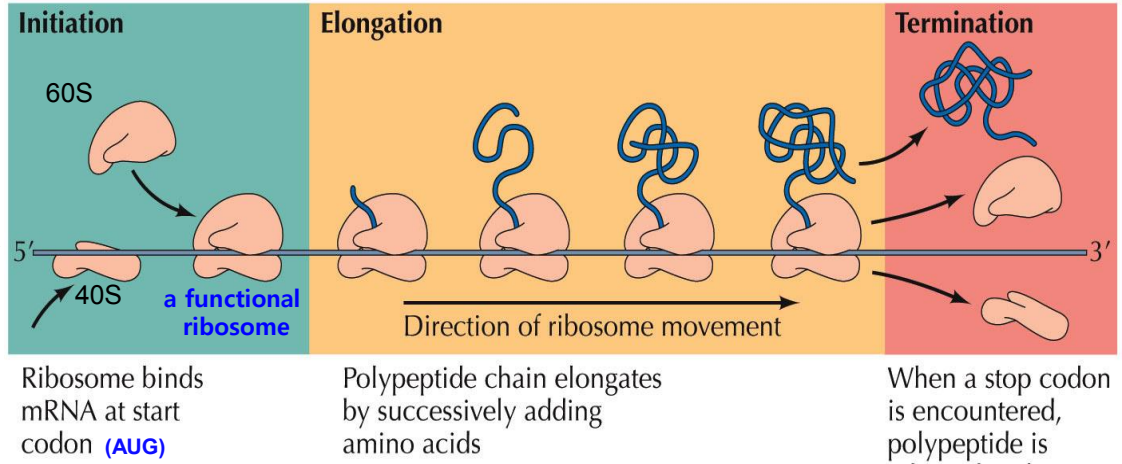



Figure 9.9. Overview of translation

## 3.2. Translation factors

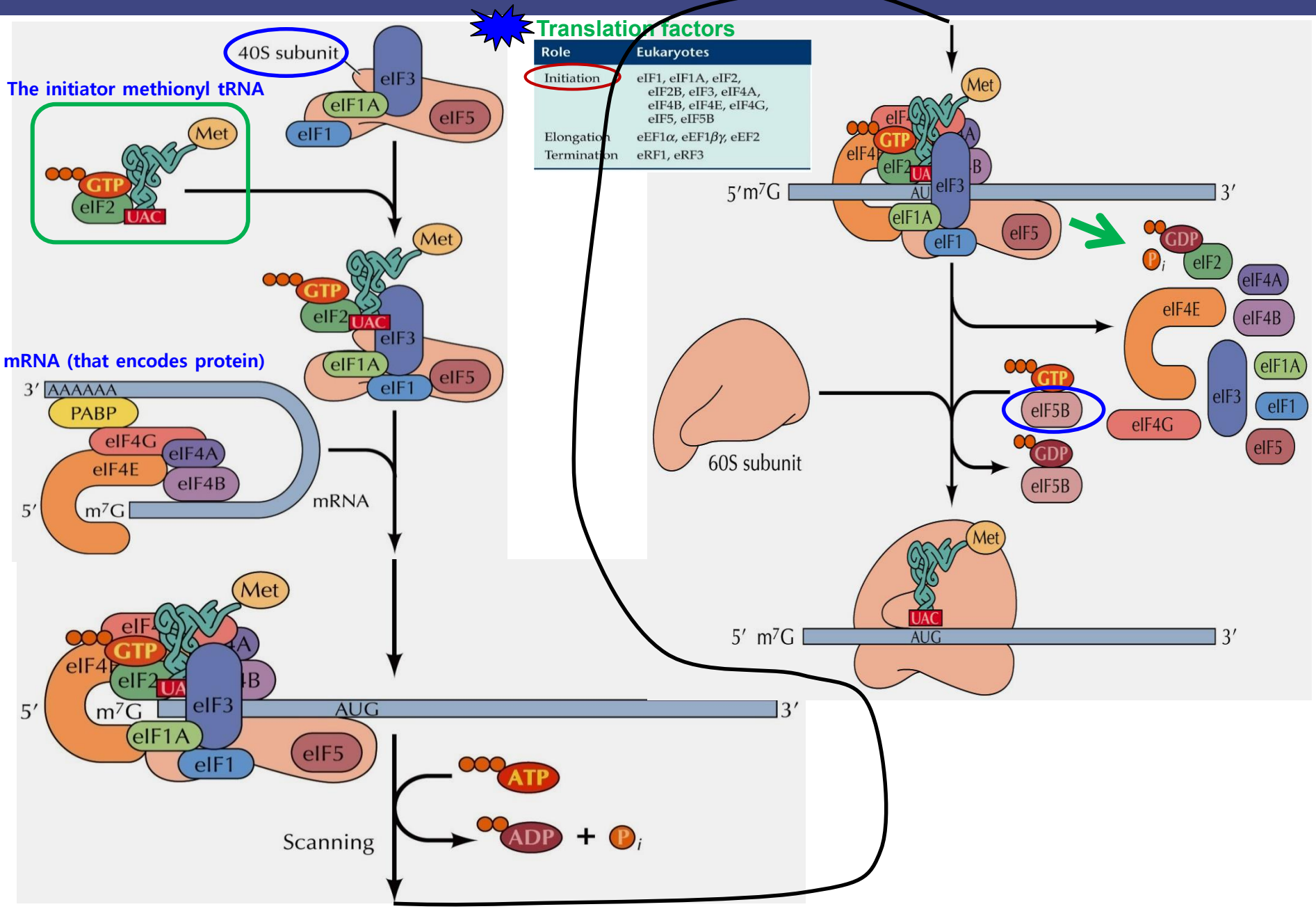
| Role | Prokaryotes | Eukaryotes |
|---|---|---|
| Initiation | IF1, IF2, IF3 | eIF1, eIF1A, eIF2, eIF2B, eIF3, eIF4A, eIF4B, eIF4E, eIF4G, eIF5, eIF5B |
| Elongation | EF-Tu, EF-Ts, EF-G | eEF1α, eEF1βγ, eEF2 |
| Termination | RF1, RF2, RF3 | eRF1, eRF3 |

기타 번역개시 관여 (비리보솜) 단백질
- PABP (Poly(A)-binding protein): mRNA의 3' poly(A) tail을 인식해 eIF4G와 상호작용 → mRNA 고리 구조 형성, 번역 효율 ↑.

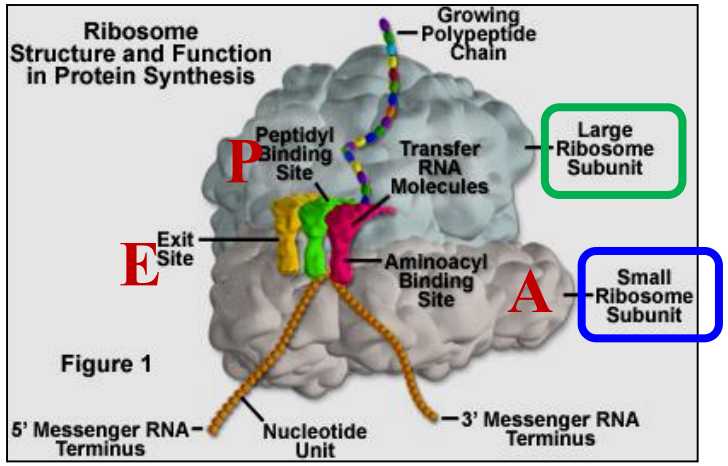
# Chapter 9. Protein Synthesis, Processing, and Regulation (1)

| 번역 단계 | 주요 단백질 | 기능 |
|---|---|---|
| **개시 (Initiation)** | **eIF4E** | mRNA 5’ cap 구조 인식 |
| | **eIF4G** | 스캐폴드 단백질, eIF4E·eIF4A·PABP 연결 |
| | **eIF4A** | ATP 의존적 helicase, 5’UTR 2차 구조 해체 |
| | **eIF2** | Met-tRNAi^Met을 40S 소단위에 전달 (GTP 결합 필요) |
| | **eIF3** | 40S 소단위 안정화, 번역 개시 복합체 형성 촉진 |
| | **PABP** | poly(A) tail 인식, eIF4G와 상호작용 → mRNA 고리 구조 형성 |
| **신장 (Elongation)** | **eEF1A** | 아미노아실 tRNA를 리보솜 A site로 운반 |
| | **eEF1B** | eEF1A의 GDP→GTP 교환 촉진 |
| | **eEF2** | 리보솜 전위(translocation) 촉진 |
| **종결 (Termination)** | **eRF1** | Stop codon 인식 → 펩타이드 사슬 방출 |
| | **eRF3** | GTPase, eRF1과 함께 종결 반응 촉진 |
| **리보솜 재활용** | **ABCE1** | ATPase, 종결 후 리보솜 소단위 분리 |
| | **Dom34/Hbs1 (Pelota/Hbs1)** | 비정상 mRNA에 걸린(stalled) 리보솜 구제 |
| **조절 (Regulation)** | **4E-BP1** | eIF4E 억제, mTORC1 신호에 의해 조절됨 |
| | **GCN2, PKR, PERK, HRI** | eIF2α 인산화 → 전반적 번역 억제 |
| | **Ago, GW182 (miRNA 복합체)** | 특정 mRNA 번역 억제 및 분해 유도 |
| | **TIA-1, G3BP 등** | 스트레스 과립 형성 → 선택적 번역 억제 |

# Figure 9.11 Initiation of translation in eukaryotic cells



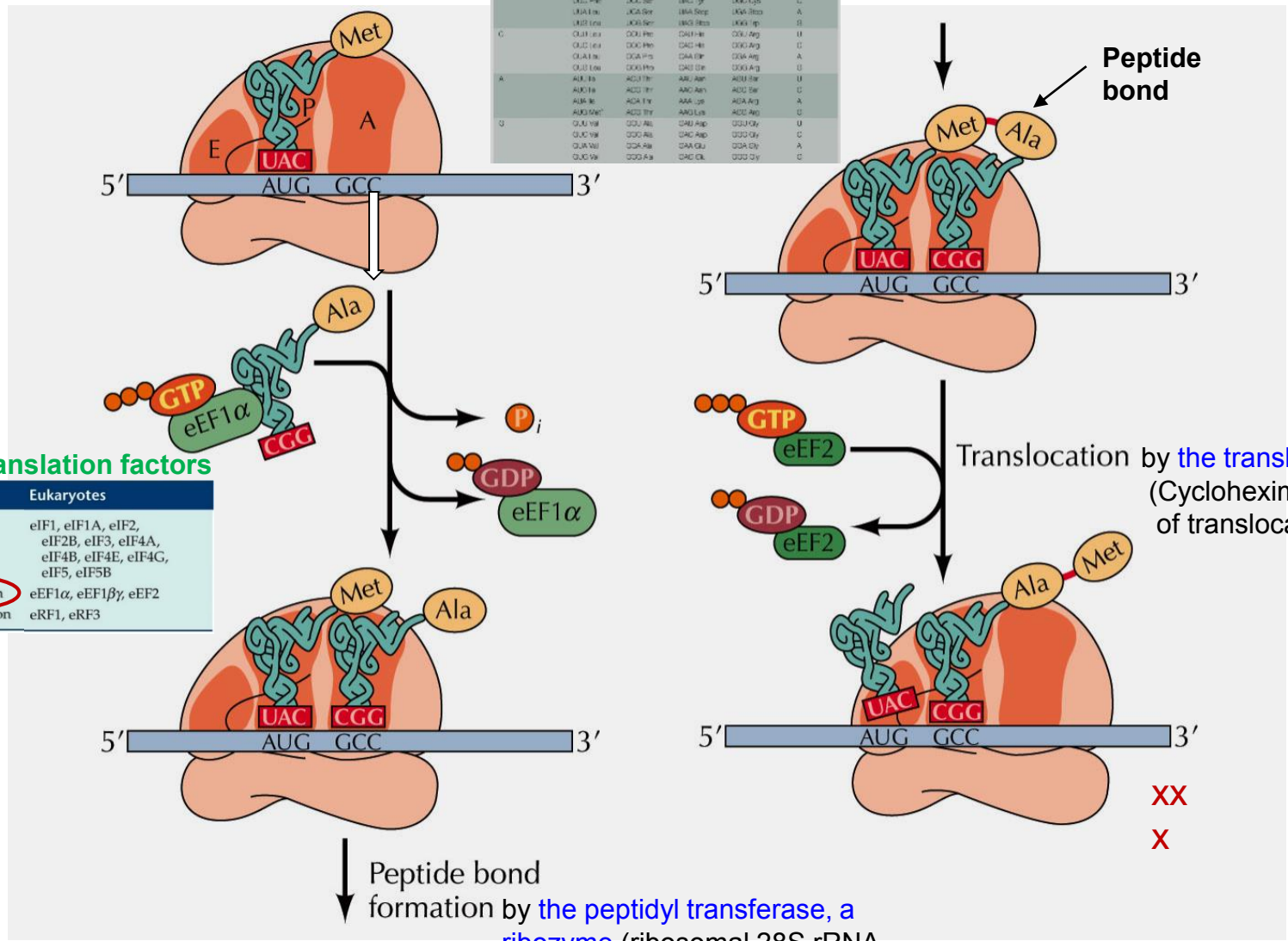
# Figure 9.13 Elongation stage of translation




▶ **The large ribosomal subunit** has **three tRNA binding sites**: **P (peptidyl), A (aminoacyl), and E (exit) sites**.

**Translation factors**

| Role | Eukaryotes |
|---|---|
| Initiation | eIF1, eIF1A, eIF2, eIF2B, eIF3, eIF4A, eIF4B, eIF4E, eIF4G, eIF5, eIF5B |
| Elongation | eEF1$\alpha$, eEF1$\beta\gamma$, eEF2 |
| Termination | eRF1, eRF3 |




XX

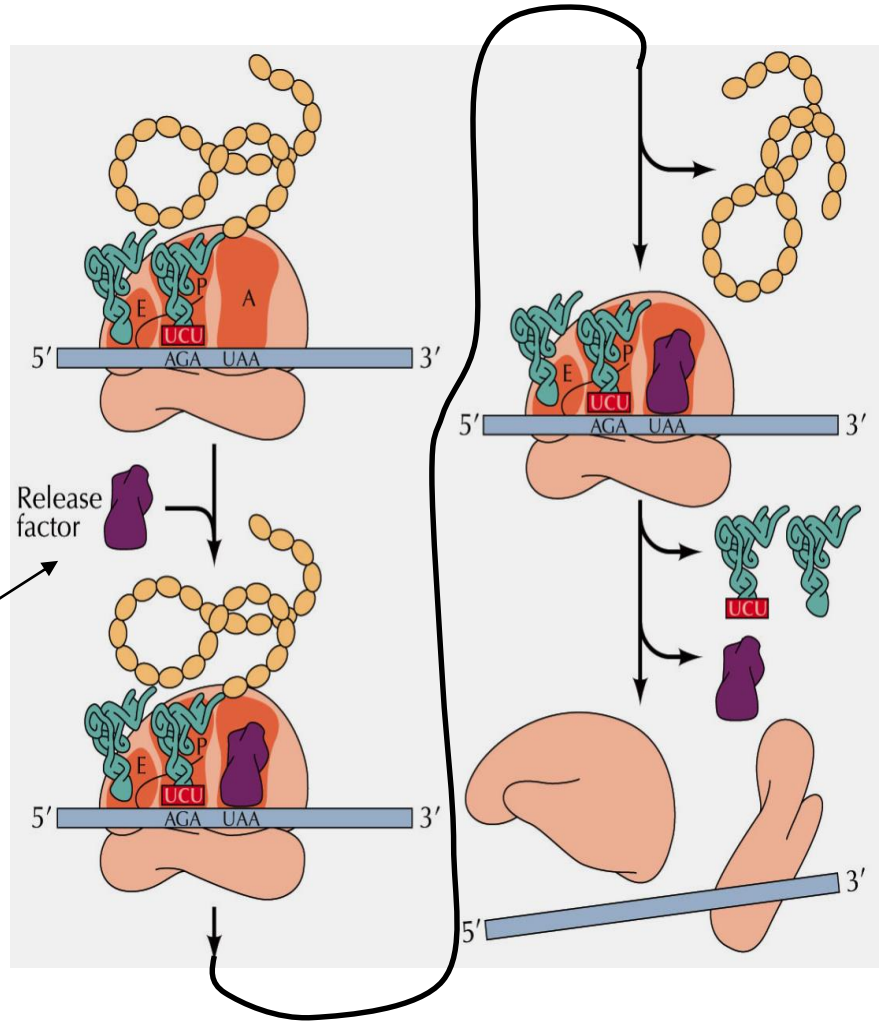
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# Figure 9.15 Termination of translation

- **Elongation continues** until **a stop codon (UAA, UAG or UGA)** is translocated into **the A site**.
- **Release factors (RF)** recognize the signals and **terminate protein synthesis**.
- In eukaryotic cells, **eRF1 recognizes all three stop codons.**

**Translation factors**

| Role | Eukaryotes |
|---|---|
| Initiation | eIF1, eIF1A, eIF2, eIF2B, eIF3, eIF4A, eIF4B, eIF4E, eIF4G, eIF5, eIF5B |
| Elongation | eEF1α, eEF1βγ, eEF2 |
| Termination | eRF1, eRF3 |



# Translation of mRNA

- **A single mRNA can be translated simultaneously by a series of multiple ribosomes in both pro- and eukaryotic cells**: Once a ribosome has moved away from the initiation site of mRNA, another ribosome can bind to the mRNA and begin synthesis of a new polypeptide chain.
- **Thus mRNAs are usually translated by a series of ribosomes, spaced at intervals of 100 to 200 nucleotides.**
- **The group of ribosomes bound to an mRNA molecule** is called **a polyribosome** (also called **polysome**).
- **Each ribosome within polysome functions independently to synthesize a separate polypeptide chain**.

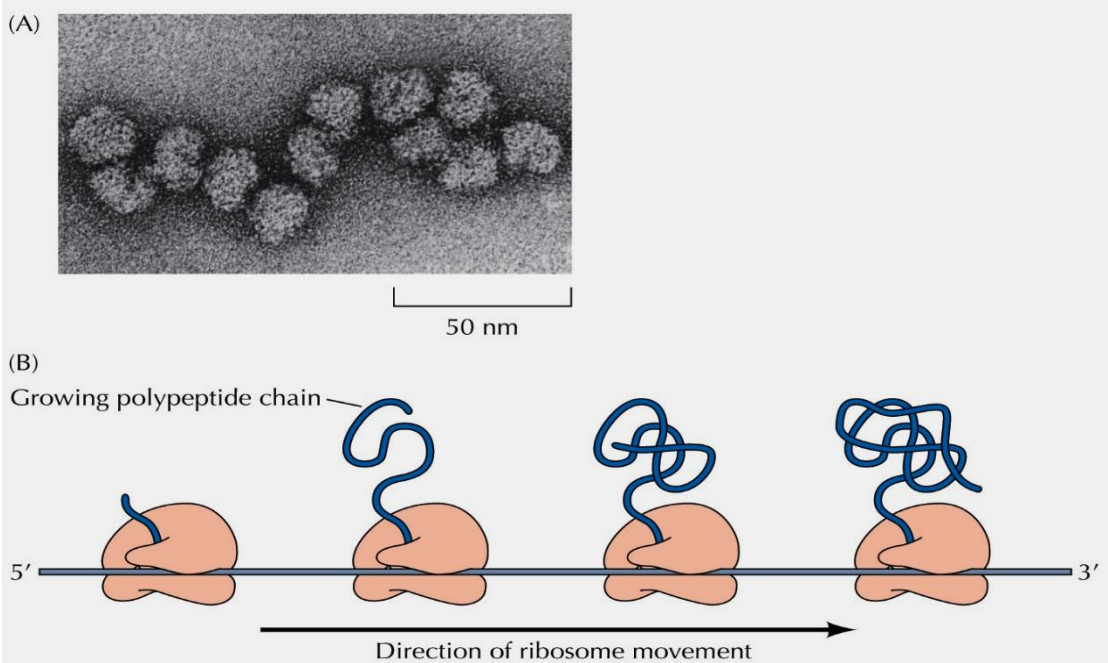



**Figure 9.16. (A) EM of a eukaryotic polysome and (B) Schematic of a generalized polysome**

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4. **Regulation of translation in eukaryotes**

# Regulation of translation

- **Regulation of translation** also plays a key role in modulating gene expression.
- Regulation of translation includes **translational repressor proteins**, **microRNAs**, and **many other cellular factors**.
- **Global translational activity of cells is modulated in response to cell stress, nutrient availability, and growth factor stimulation**.

▶ **Translation of a ferritin mRNA** encoding **a ferritin protein that stores iron in cells** is regulated by **repressor proteins**, **depending on cellular levels of iron**.

▶ **When iron is absent**, **iron regulatory protein (IRP)** in cells binds to **the iron response element (IRE) in the 5′ UTR**, thereby **blocking translation**.

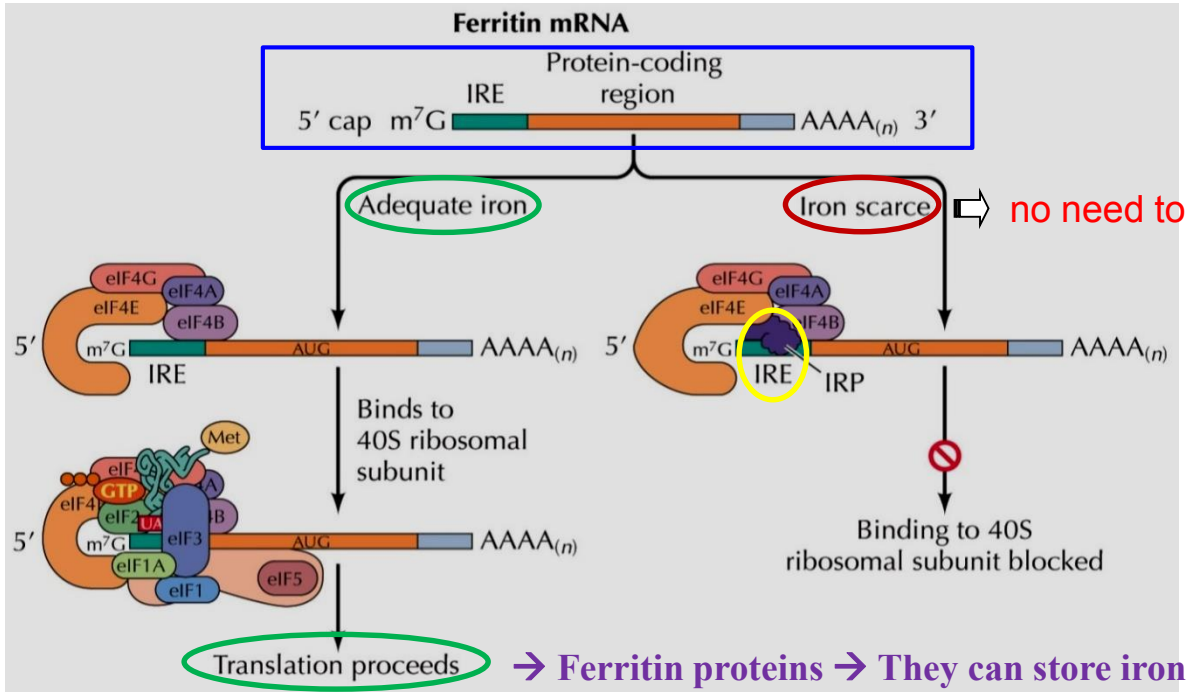



**Figure 9.17. Translational regulation of ferritin**

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다음주 9/28(월)

# Chapter 9
# Protein Synthesis, Processing, and Regulation 2

**Thanks for your attention!**